higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PULP AND PAPERMAKING TECHNOLOGY
NQF LEVEL 3

4 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 False
1.2 True
1.3 True
1.4 True
1.5 True

(5 × 1) **[5]**

QUESTION 2

2.1 Contraries
2.2 Deflaker
2.3 Density
2.4 Mathieson process
2.5 Nip

(5 × 1) **[5]**

QUESTION 3

3.1 F
3.2 D
3.3 A
3.4 I
3.5 B

(5 × 1) **[5]**

QUESTION 4

4.1
- The correct length of the logs (2,4 m)
- The thickness of the logs
- It the logs are still fresh or if they are old and rotting
- Whether the logs have cracks in them

(Any 3 × 1) (3)

4.2 4.2.1
- Water protects the logs so that they last longer
- Barks are removed easier from wet logs than dry logs

4.2.2
- Keeping logs in large pools of water
- Spraying water onto the logs

(2 × 2) (4)

4.3 The first logs that are stacked from the storage will be the first logs that are taken out to have their bark removed or to debarking. (1)

4.4
- If there is still some bark left on the logs that have come out of the debarker.
- If the debarker is not removing bark from as many logs as it should in a day.
- If the debarker is removing wood along with the bark. (3)

4.5 4.5.1 Change the knife that has become dull with sharpened knife.

4.5.2 Wood should be kept moist so that the chips of the correct size and quality can be cut.

(2 × 1) (2)

4.6
- The feed to the press should be adjusted
- Screen should be checked for blockage (2)

4.7
- To avoid contraries from breaking the machinery and preventing low quality pulp from being produced
- Contraries can affect the quality of the bagasse and if there are stones or metal pieces, they can damage the equipment. (2)

[17]

QUESTION 5

5.1
A Pulp
B Stator
C Deflaker pulp
D Deflaker plates
E Rotor (5 × 1) (5)

5.2 5.2.1 Viscometer

5.2.2 Conductivity meter

5.2.3 Visual inspection using reflection light

(3 × 1) (3)

5.3 It is used for paper with large areas of ink on it.✓ Surfactants like detergents, dispersant and foaming agents are added to the pulper.✓ Foaming agents are added to cause the water-ink mixture to float on the surface of the pulper slurry.✓ The pulp is then sent to the ink floatation cells to remove the ink.✓ In the floatation cell, air is pumped into the slurry.✓ The ink and dirt particles stick to the air bubbles and rise to the top and this causes a layer of foam to form on the surface of the floatation cell.✓ The foam layer is made up of ink, dirt and air bubbles.✓ (7)

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5.4 5.4.1 Shower water flow must be increased and showers checked for blockages.

5.4.2 Stone must be sharpened.

(2 × 1) (2)
[17]

QUESTION 6

6.1 6.1.1 The wood chips were not steamed long enough.

6.1.2 Temperature in the pre-steamer is too high.

6.1.3 Temperature in the steamer is too high which liquefies the lignin in the chips.

(3 × 1) (3)

6.2
- Stiffer and stronger fibres than mechanical methods
- A higher yield than fully chemical pulping. But with the same quality fibre
- The pulp produced makes paper with good printing properties
- High pulp yield of 55%–85% (Any 3 × 1) (3)

6.3
- Vertical batch digester
- Vertical continuous digester
- Horizontal continuous digester (3)

6.4

Kraft pulping process	Acid pulping process
Digester is made of cheap mild steel	Digester is made of expensive stainless steel to avoid corrosion
Used cooking liquor leaving the digester is called black liquor	Used cooking liquor leaving the digester is called red liquor
Cooking liquor is an alkali	Cooking liquor is an acid

(Any 2 × 2) (4)

6.5
- Toilet paper
- Serviette paper
- Paper towels
- Towels
- Industrial wipes (Any 4 x 1) (4)

[17]

QUESTION 7

7.1 7.1.1

- Remove the lignin
- Increase the value and quality of the pulp by improving appearance of the pulp product
- Remove visible traces of ink from secondary fibres increase the value of the pulp
- Breakdown lignin and increase solubility (Any 3 × 1) (3)

7.1.2

- Chlorine
- Oxygen
- Hypochloride
- Chlorine dioxide
- Hydrogen dioxide
- Ozone (Any 4 × 1) (4)

7.2 The raw materials are sodium chloride (NaCl), water and hydrochloric acid (HCl).✓ Sodium chlorate is made using the NaCL and water.✓ Sodium chlorate is reacted with HCl to form ClO_2.✓ The ClO_2 gas is then absorbed into chilled water and stored.✓ (4)

7.3

- Rotary vacuum washers
- Atmospheric diffusion washer
- Pressure diffusion washer
- Kamyr digester washer
- Horizontal belt washer (5)

7.4 Too high flow of the washer (1)

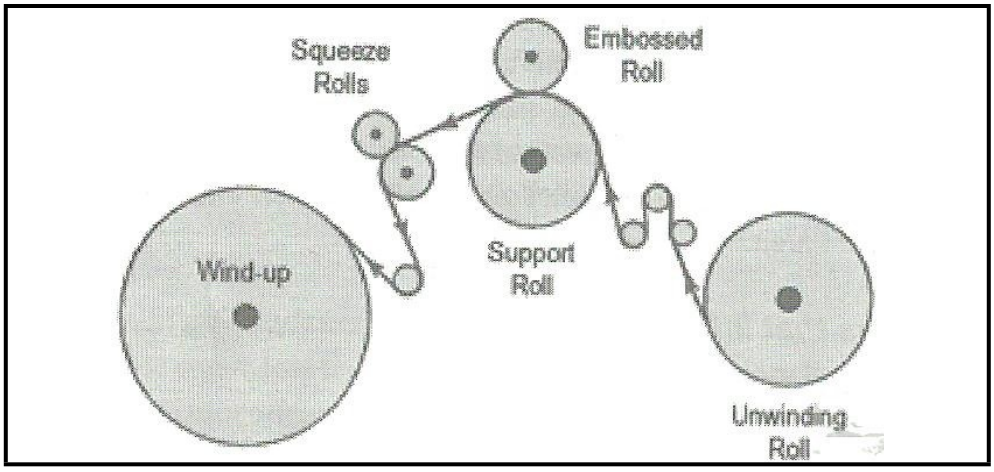
[17]

QUESTION 8

8.1 8.1.1

- To decorate
- To increase the quality of paper (2)

8.1.2



ONE MARK FOR EACH CORRECT LABELLING (5)

8.2 8.2.1 Spread the web out and flatten it to make sure that there are no crease before glue is applied.

8.2.2 Smooth the paper after lamination and control the thickness of the paper.

(2 × 2) (4)

8.3 If there are holes in the sheet before coating. (1)

8.4 Agitators are devices in tanks that keep the slurry well mixed and make sure that the solids do not settle on the bottom of the tank. (2)

8.5 By checking and resetting the scale. (1)

8.6 8.6.1 When the top of the pile of paper formed is kept at almost same height as the tape along which cut paper is transported.

8.6.2 When the paper stacks formed are vibrated so that the edges of all the paper in the stack line up.

(2 × 2) (4)
[17]

TOTAL: 100

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